



Hot Chips Tutorial Part III

RISC-V CPUs in NVIDIA Servers

Frans Sijstermans, NVIDIA

Hot Chips | August 2026



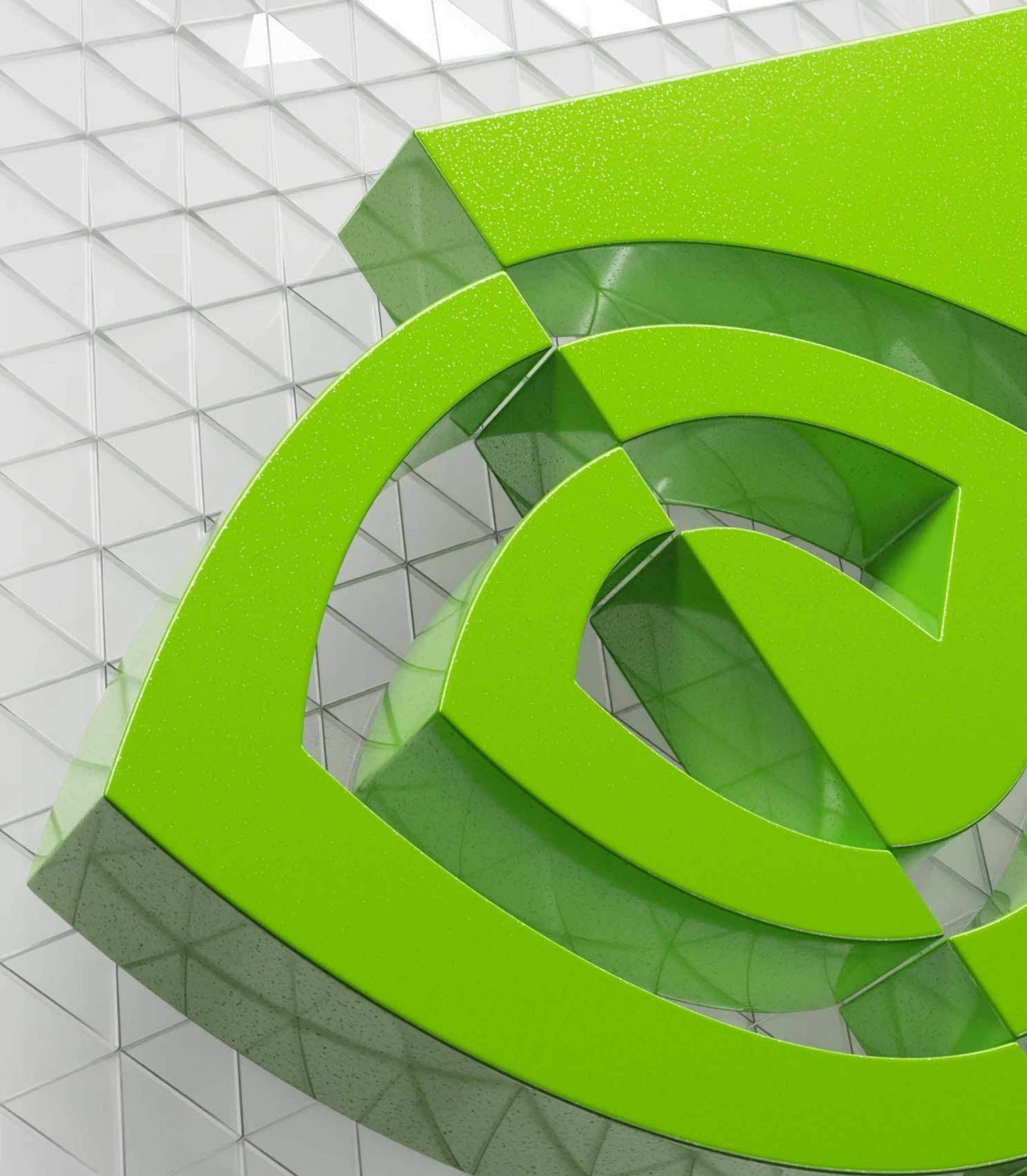
Agenda

Two platforms: CUDA and NVLink Fusion

Role of the CPU in the platforms

Requirements for RISC-V CPUs

Opportunities and challenges



CUDA – 20 years of accelerated computing

Compute Unified Device Architecture

Introduced in 2006, CUDA is a parallel computing platform and programming model that allows developers to use GPUs for general-purpose processing



Enables **General Purpose GPU (GPGPU)** computing



The foundation for all modern AI and deep learning

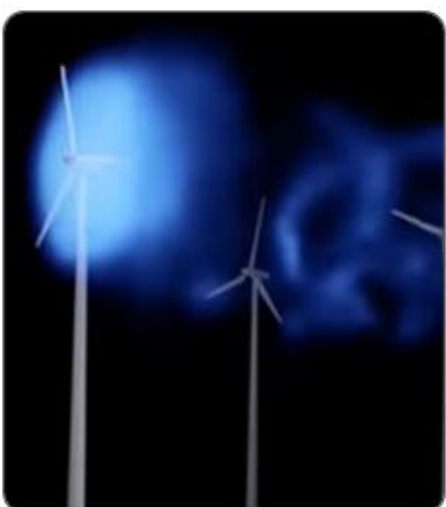


20 years of continuous hardware-software co-evolution

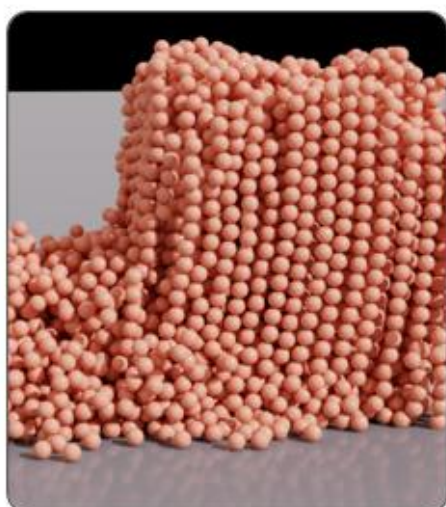
CUDA-X Accelerates Every Industry



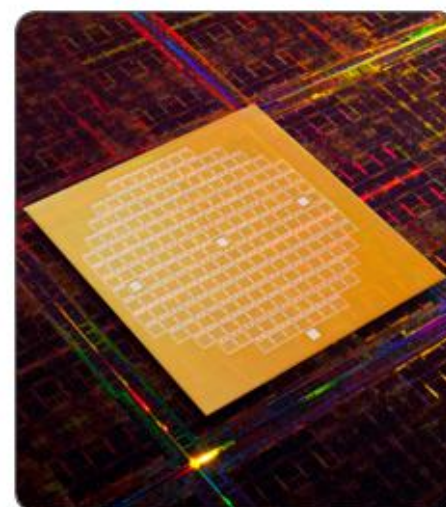
cuDSS
CAE



PhysicsNeMo
AI Physics



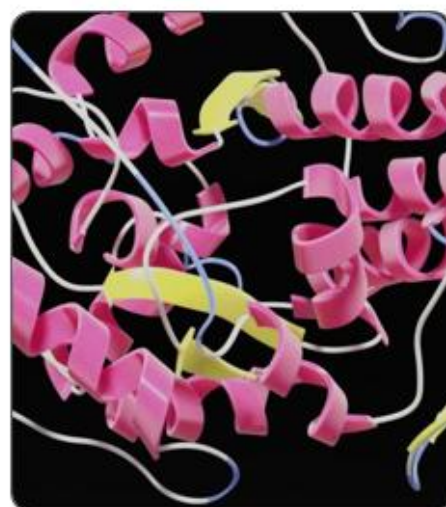
Warp
Physical Simulation



cuLitho
Computational
Lithography



ALCHEMI
AI Materials Science



cuEquivariance
Drug & Materials
Discovery



Parabricks
Gene Sequencing



CUDA-Q
Quantum Computing



cuOpt
Decision Optimization



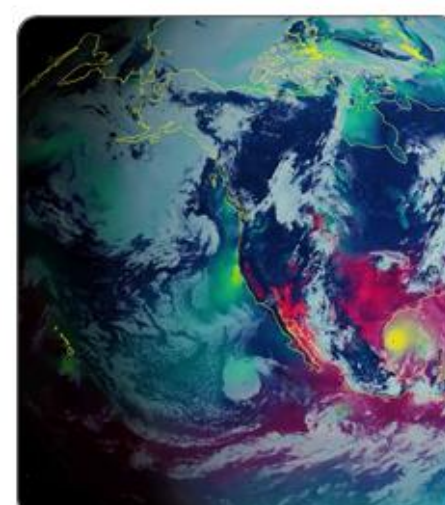
cuDF
Data Processing



cuPyNumeric
Numerical Computing

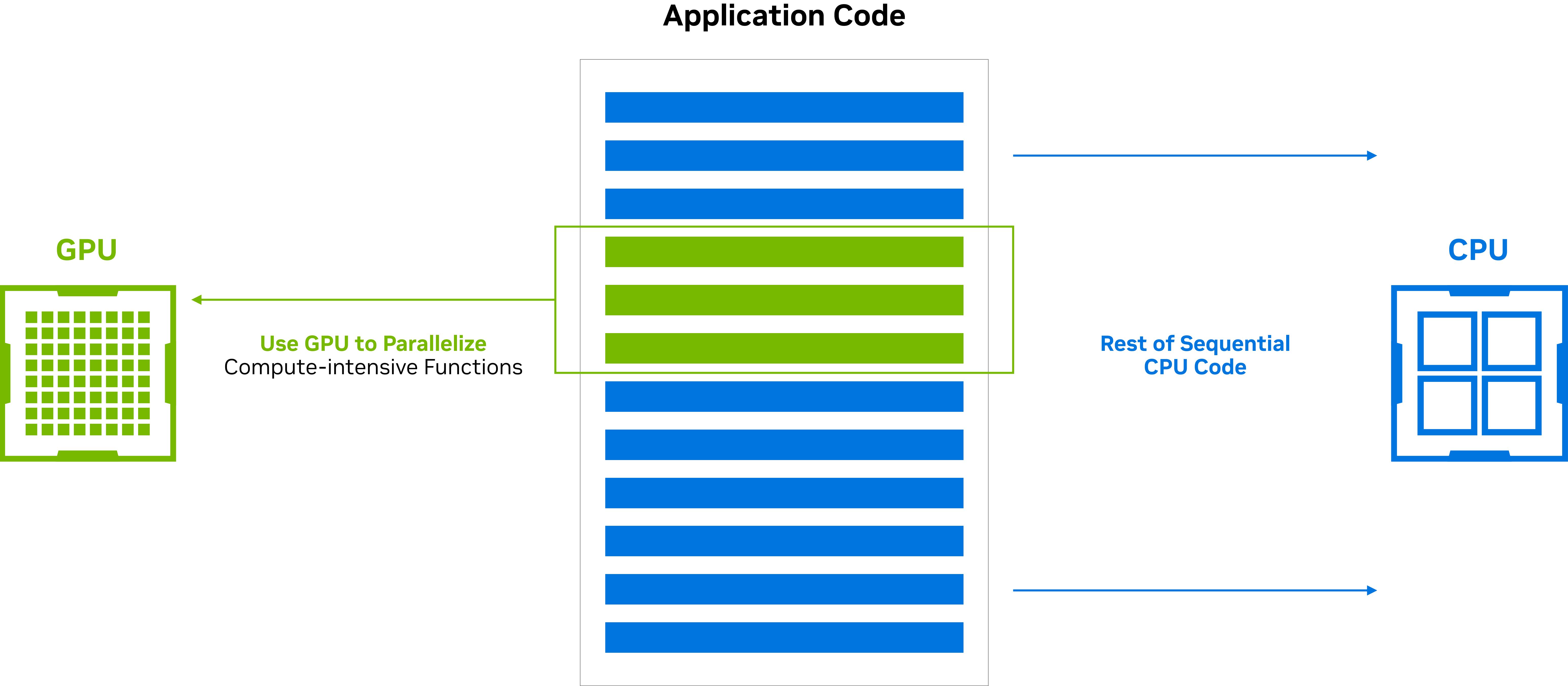


Holoscan
Edge HPC



Earth-2
Weather Analytics

CUDA applications combine CPU and GPU SW modules



Host CPU SW is complex and heterogeneous

Application SW on the CPU consist of:

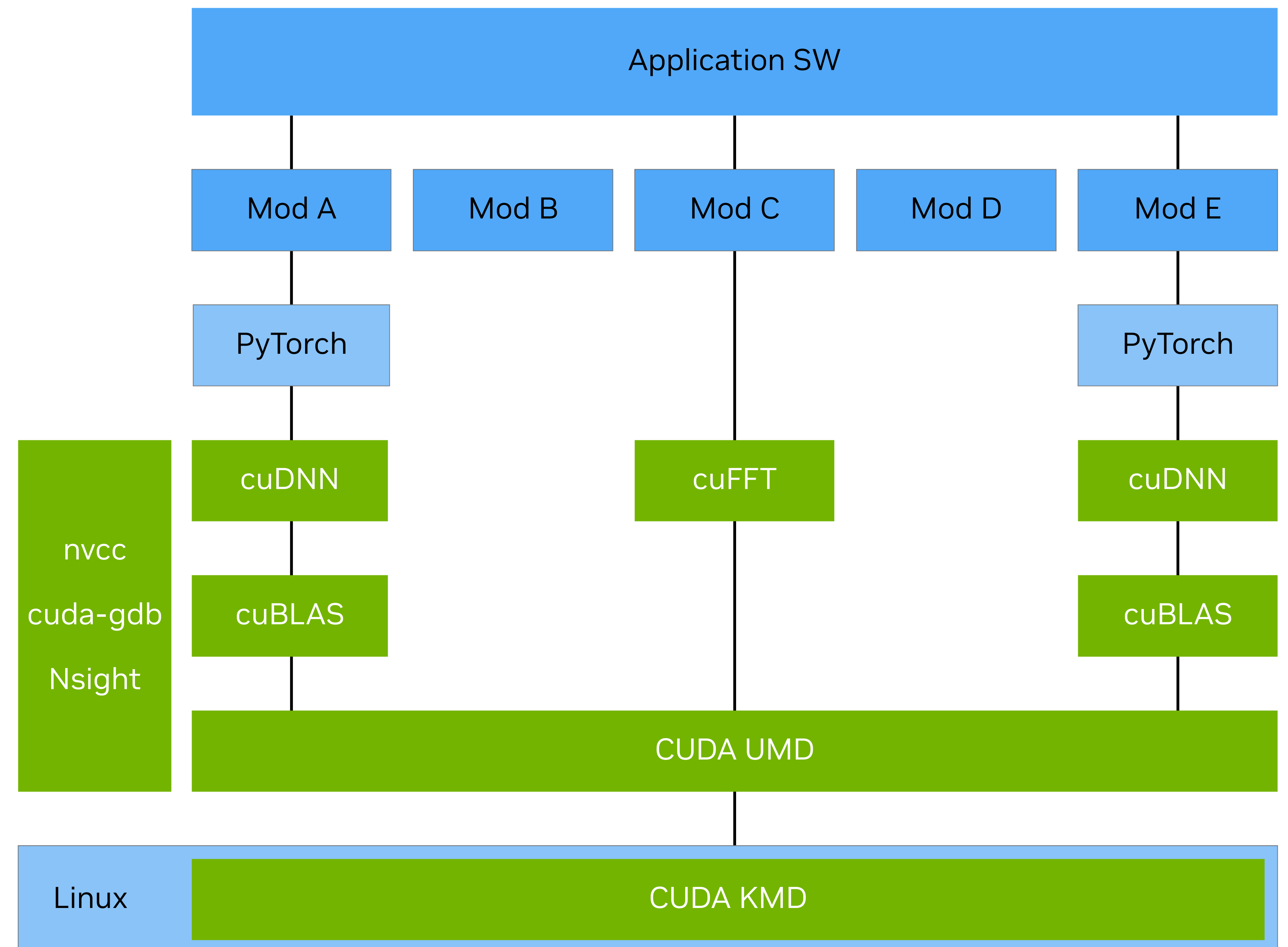
Application modules

- CPU only
- CUDA accelerated

CUDA components

- CUDA libraries
- CUDA drivers
- CUDA tools, e.g., compiler, debug

Third party software to the target CPU



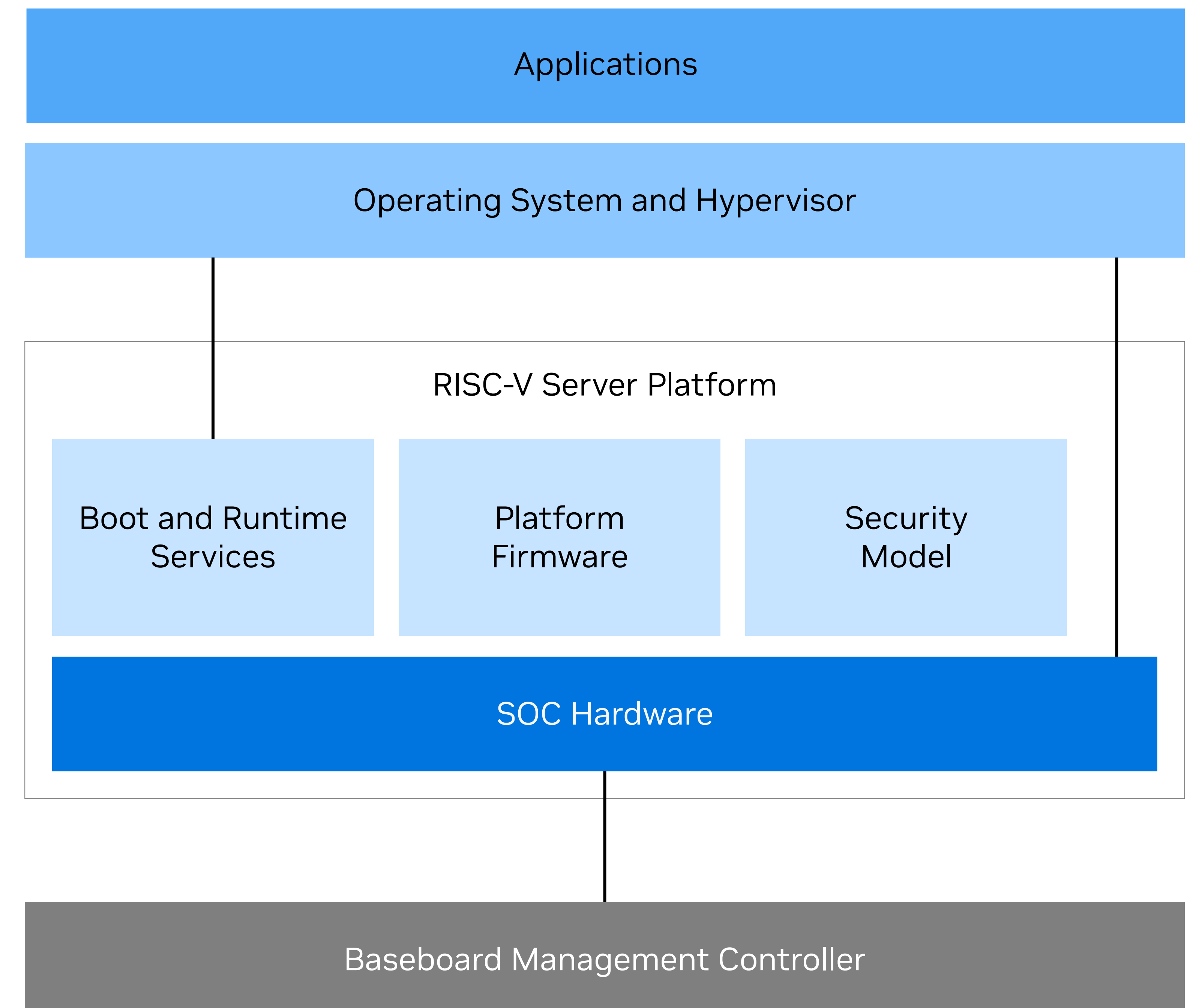
CUDA requirements for RISC-V CPUs

Adhere to RISC-V Server-related specifications

- RVA23 Profiles
- RISC-V Boot-and-Runtime Services Specification (BRS)
- RISC-V Server SOC Specification
- RISC-V Server Platform Specification

Short list of CUDA specific requirements

- Mostly related to data movement between memories
- Requirement document is available to partners



RISC-V Server specifications enable the SW ecosystem

Software binary compatibility

- RVA23 profile guarantees key extensions are present
- SW will work over a variety of RISC-V processors

Competitive performance features

- E.g. vector(V) extension, bit-manipulation(B) extension
- Optimization without code branches
- No problematic 'lowest common denominator'

Key SOC components

timers, interrupt controller, PCIe, security features, ...

Provisions for optional features

- Additional HW features are discoverable
- Accessible via kernel patches or code branches

ACPI – an example of a RISC-V platform requirement

Advanced configuration and power interface specification

Identified as critical at start of CUDA port

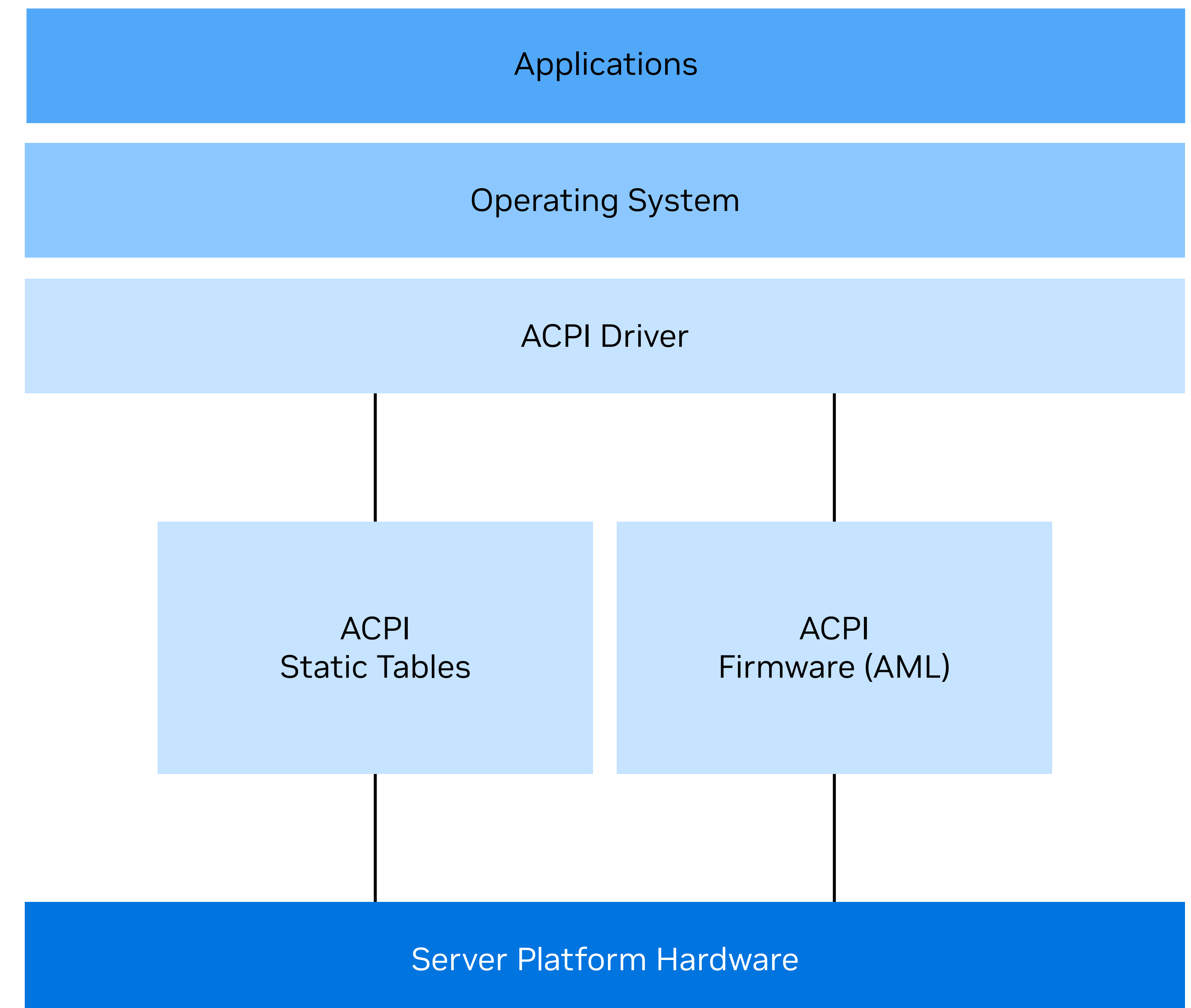
UEFI Forum added RISC-V support in ACPI 6.6 spec, ratified May 2025

RISC-V BRS spec, ratified August 2025 by RVI, specifies the ACPI requirements

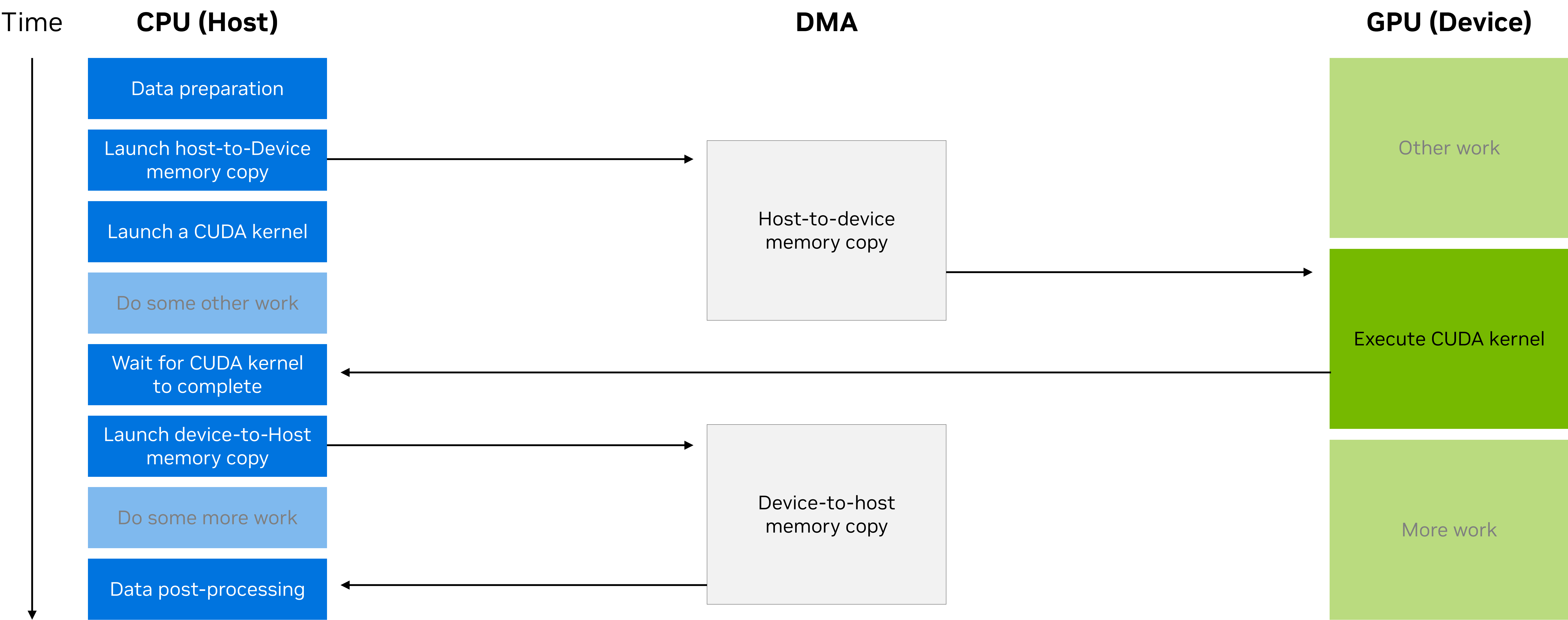
ACPI decouples OS development from HW development

Used by for many functions

- Discovery
 - CPU capability discovery (RISC-V ISA string)
 - Interrupt controllers, timers, ...
 - PCI, devices, ...
- Power, perf, thermal management
- ...



CUDA applications rely on data transfer between memories



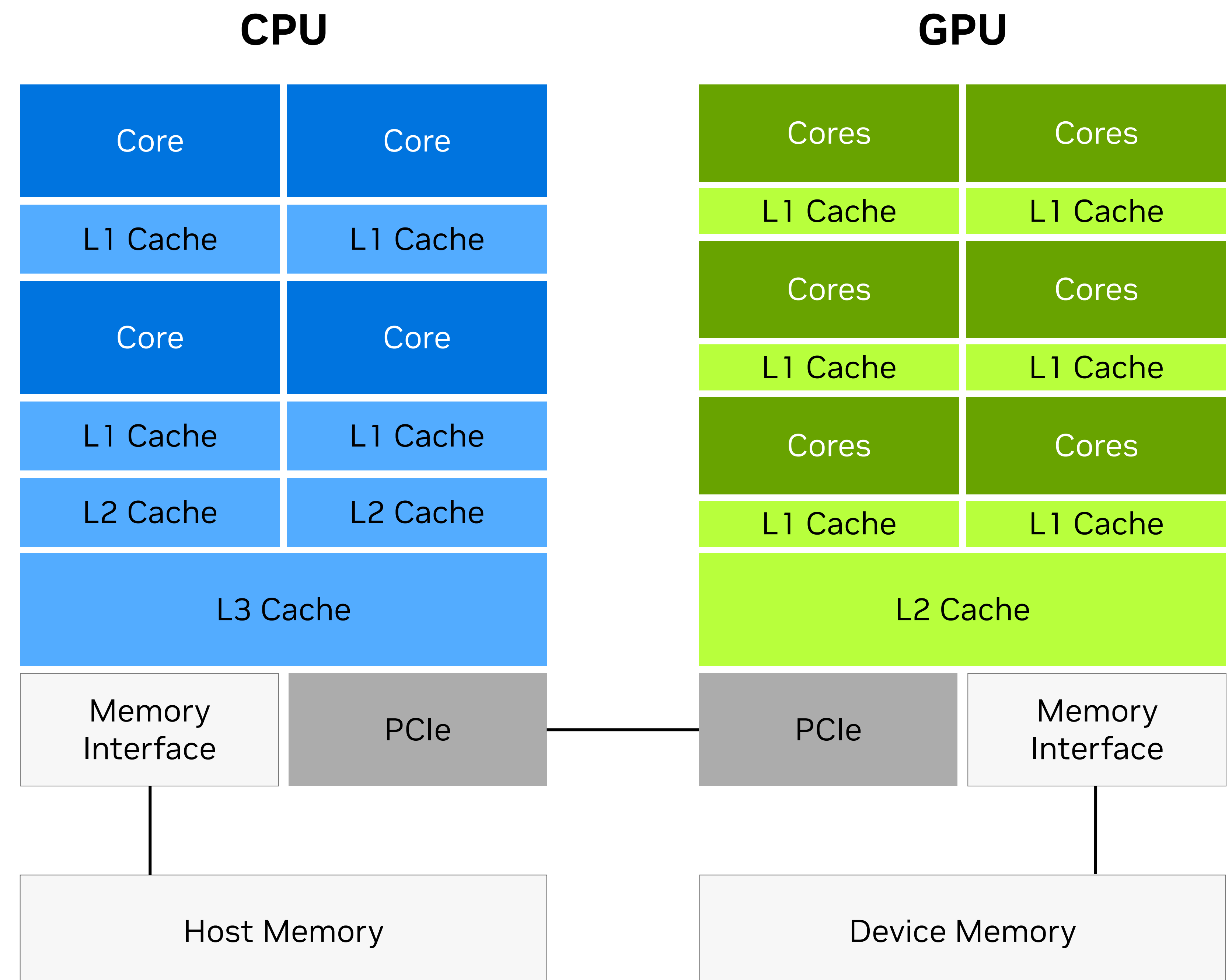
PCIe coherency – an example of a CUDA specific requirement

Host-to-device and device-to-host memory copies are common in CUDA software

HW PCIe I/O coherence avoids frequent cache flush and invalidate operations

- Simplifies software
- Improves latency
- Offloads CPU

Supported in most (all?) other modern server CPUs



PCIe P2P communication – another CUDA specific requirement

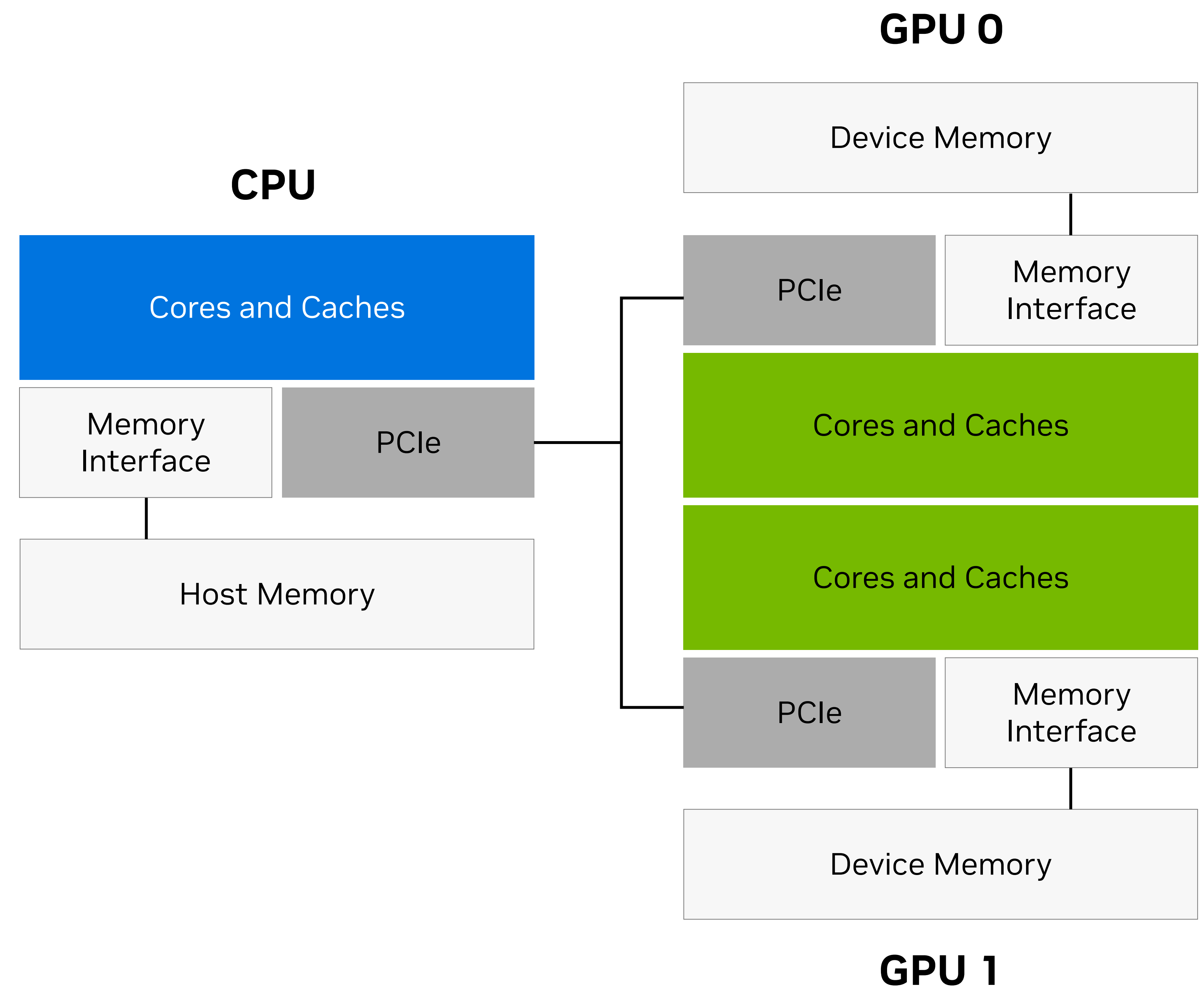
In multi-GPU systems, memory copies between two device memories are common.

Example:

1. GPU 0 writes data to GPU 1 device memory
2. GPU 0 writes flags to host memory
3. CPU reads flags from host memory
4. CPU launches CUDA kernel on GPU 1

PCIe P2P support is required for CUDA—avoids CPU involvement and extra copy to host memory

- Various root-complex and end-point configs
- P2P traffic at full speed for 128-byte packets
- Root complex to maintain standard PCIe ordering rules



NVIDIA Vera Rubin Is a Full-Stack AI Factory Platform

Extreme co-design for agentic AI across seven chips and five racks

Vera Rubin
NVL72



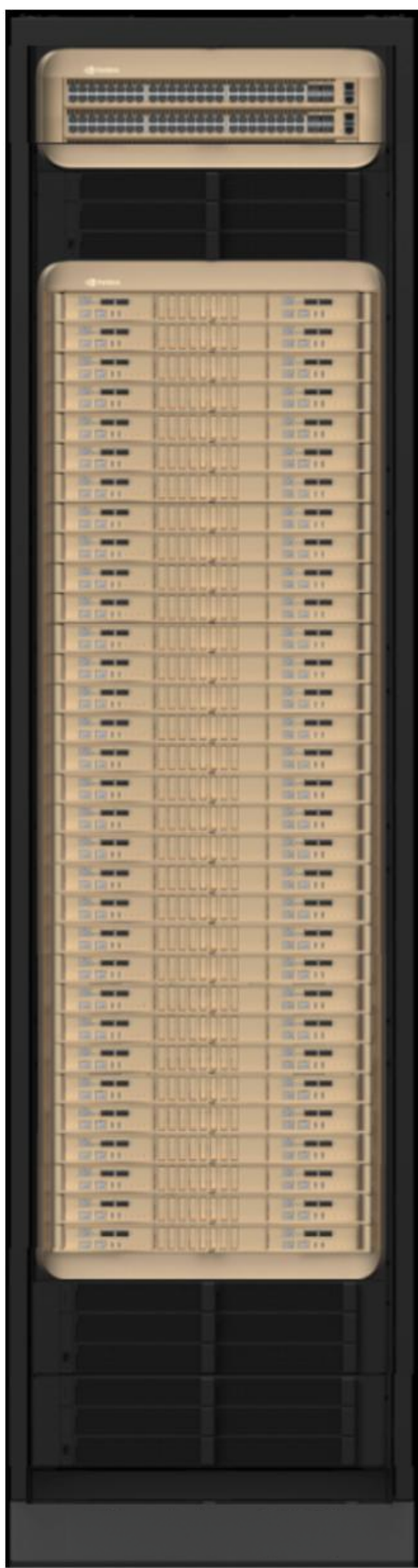
Foundation Platform

Groq 3
LPX



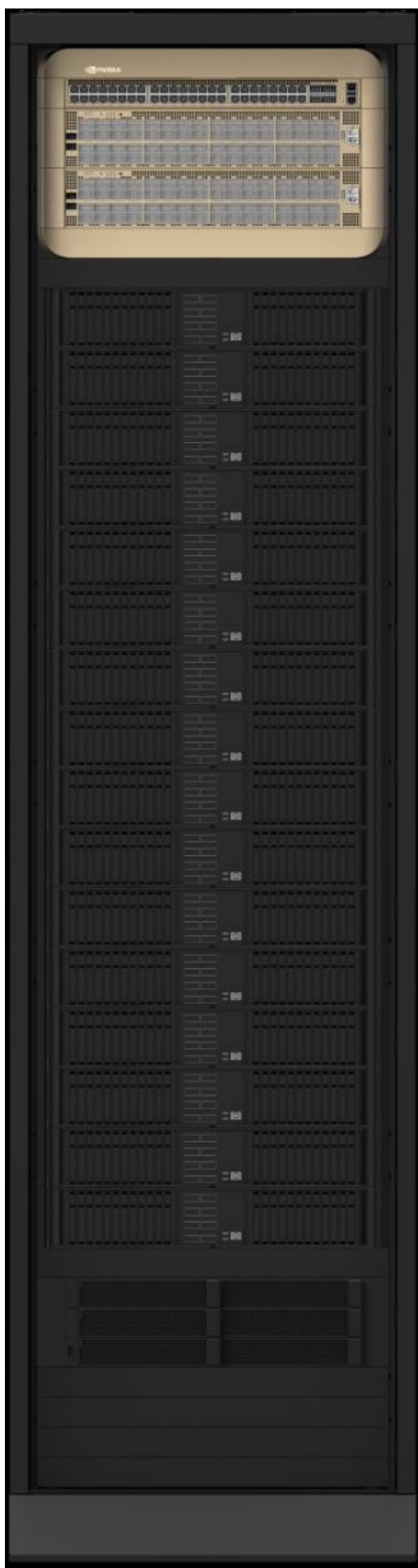
Extend Interactivity

Vera CPU
Rack



Tool Calls and Sandboxes

Vera BlueField-4
STX Storage



Context Memory Storage

Spectrum-6 SPX
Networking



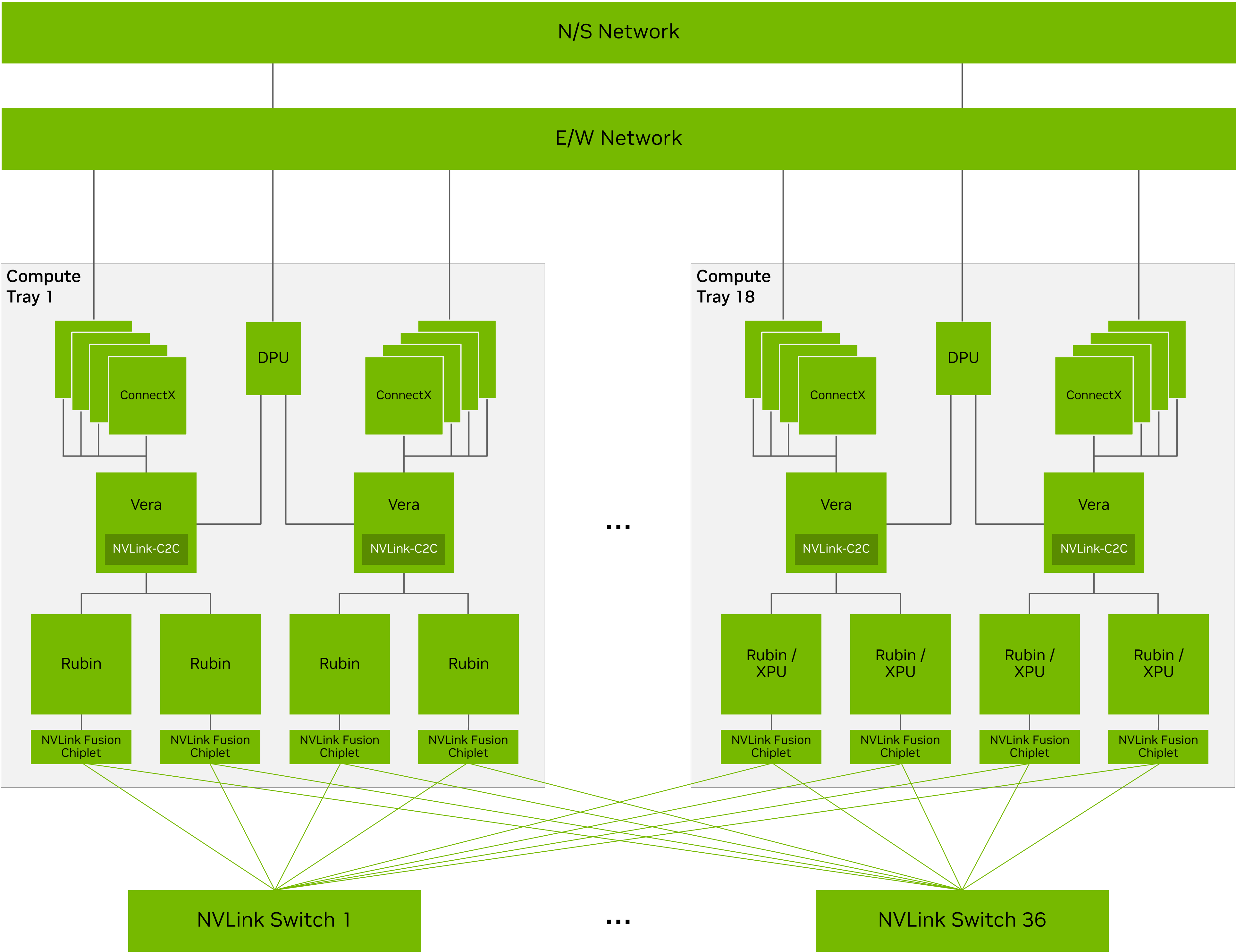
Scale-out Fabric

NVIDIA NVL72 Rack

- OCP MGX Rack**
- Single 72 GPU L1 Domain**
Fully copper domain
- 9X Switch Trays**
4X NVLink 6 switches per tray
28.8 TB/s per switch tray
- 18X Compute Trays**
4X Rubin GPUs per tray
2X Vera CPUs per tray
3.6 TB/s per GPU
900 GB/s CPU to GPU



Vera Rubin NVL72



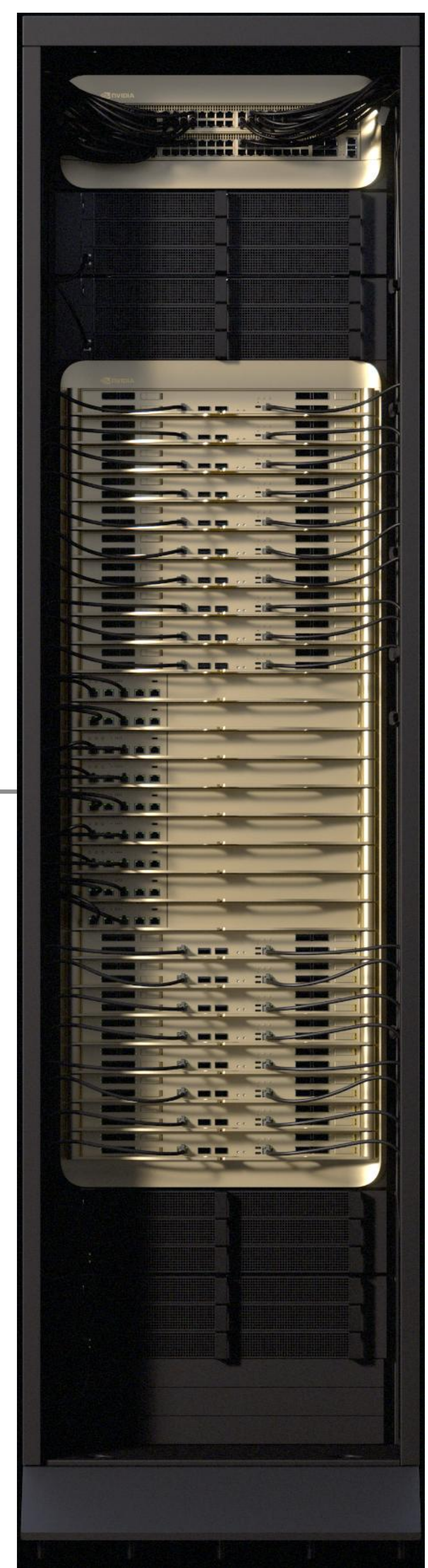
NVLink Fusion Connects Custom Silicon to NVIDIA AI Platform

OCP MGX Rack

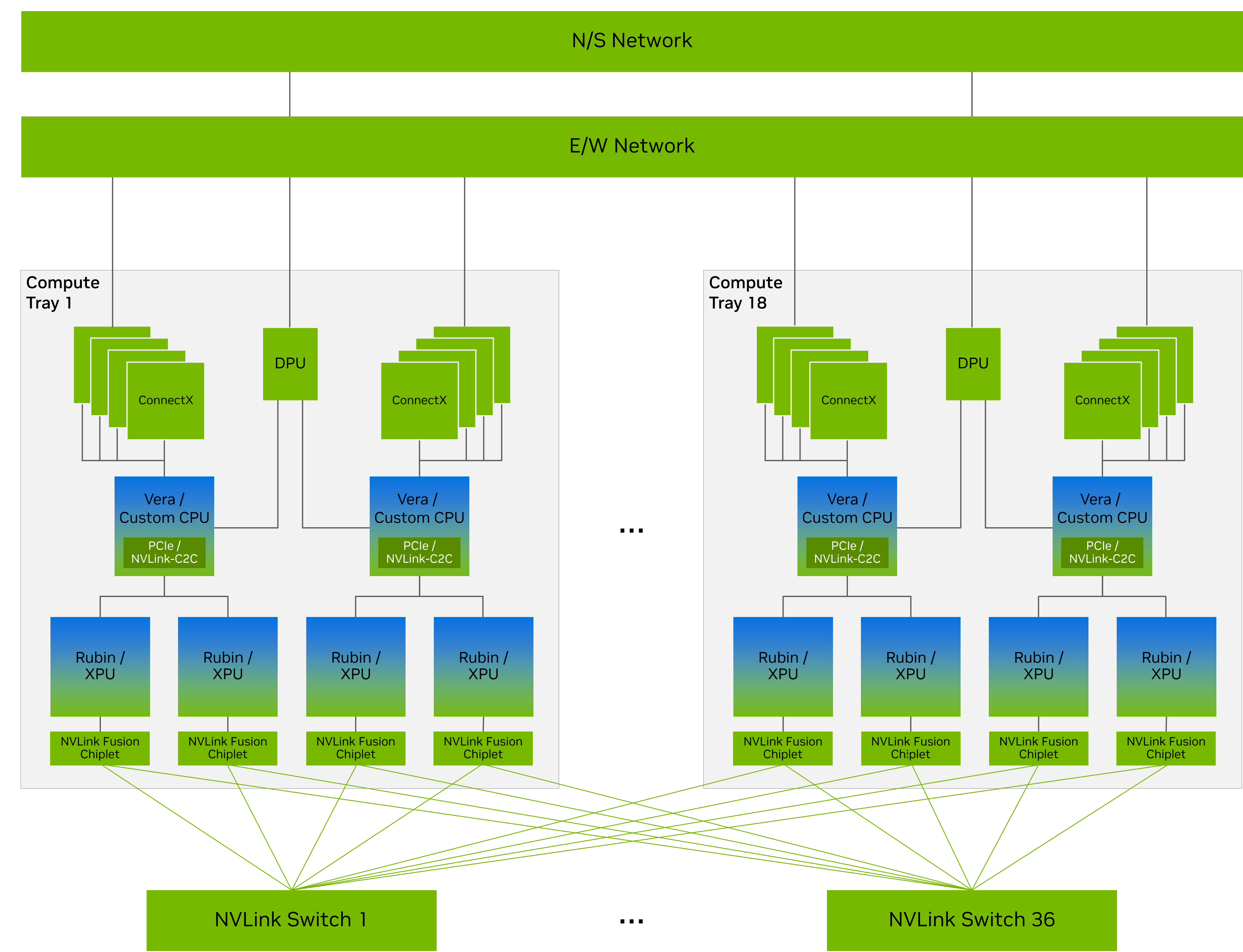
Single 72 XPU L1 Domain
Fully copper domain

9X Switch Trays
4X NVLink 6 switches per tray
28.8 TB/s per switch tray

18X Compute Trays
4X GPUs/XPUs per tray
2X CPUs per tray
3.6 TB/s per XPU
900 GB/s CPU to XPU



Vera Rubin NVL72

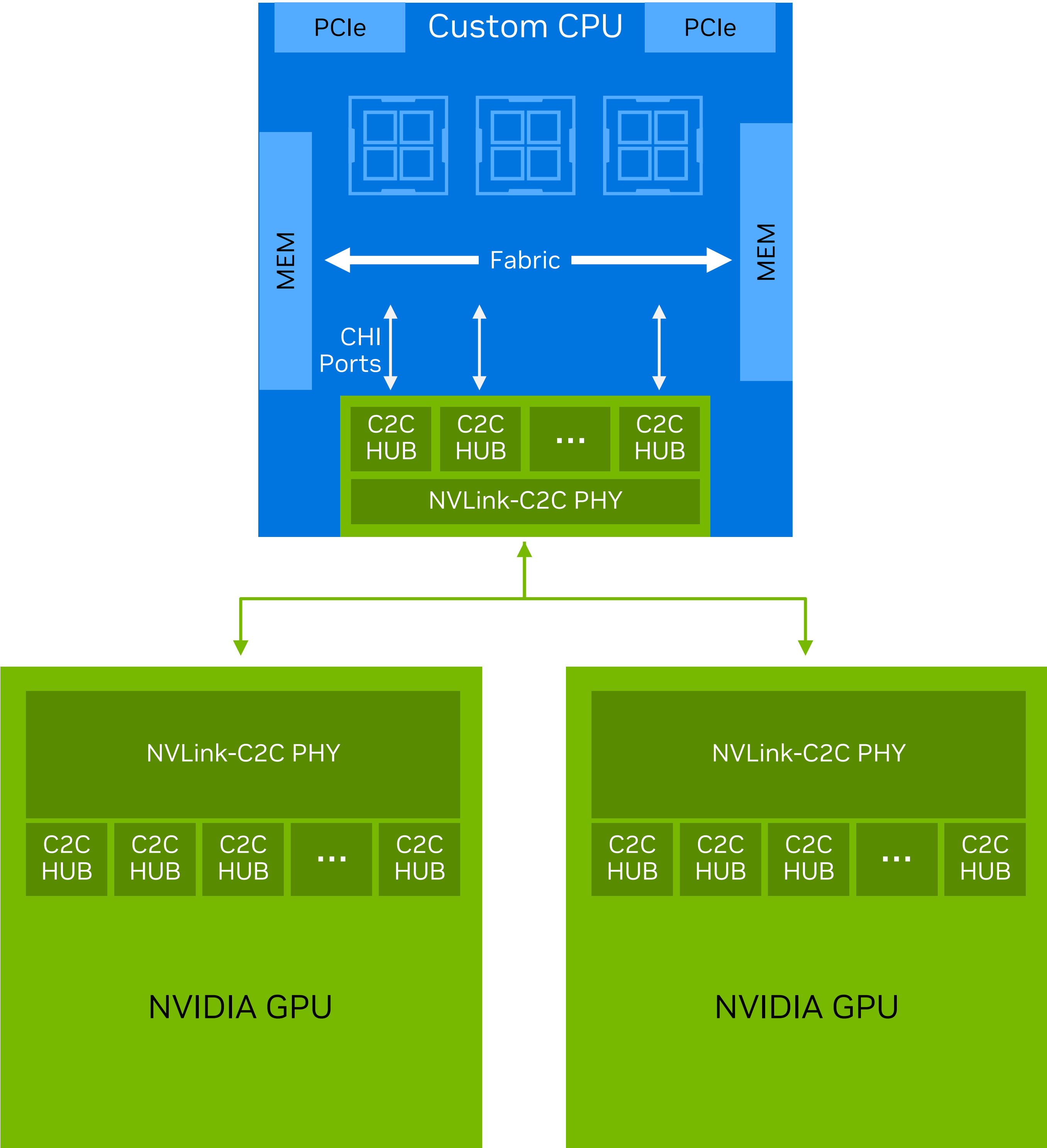


C2C interface integration in Custom CPU

NVLink IP Integration
Soft IP + PHY

CPU – GPU High Bandwidth Interface

Unified Memory Architecture
GPU and CPU can access all memory
Coherence protocol (CHI)



NVLink Fusion requirements for custom CPUs

Requirements

- All requirements for CUDA
- High-speed interconnect: C2C, ~88 PCIe lanes
- Even more software e.g.:
 - DOCA – networking
 - NCCL – multi-GPU communication
- Close partnership with NVIDIA

Example CPU specifications:

- SPECrate 2026 integer: 400
- Memory: 1 TB/s, 1 GB capacity
- Cache: L1 64+64kB, L2 1MB, L3 100MB
- ...

but *system level performance and power for targeted workload* matter more than raw specs



RISC-V broadens Server CPU choice alongside x86 and Arm

Advantages of RISC-V for server customers

- Open standard
- Multiple vendors
- Standardization combined with customization -
platforms/profiles combined with optional standard/proprietary extensions

Advantages for RISC-V developers to work with NVIDIA/platform producer

- Mature HW platform and SW stack
- NVIDIA has plans for RISC-V in its platforms
- *Don't boil the ocean!*

Growing ecosystem readiness

- RISC-V International standards ready to develop server HW and SW
- RVA23 compliant CPUs are now available
- RISC-V Software Ecosystem (RISE) project drives SW porting

